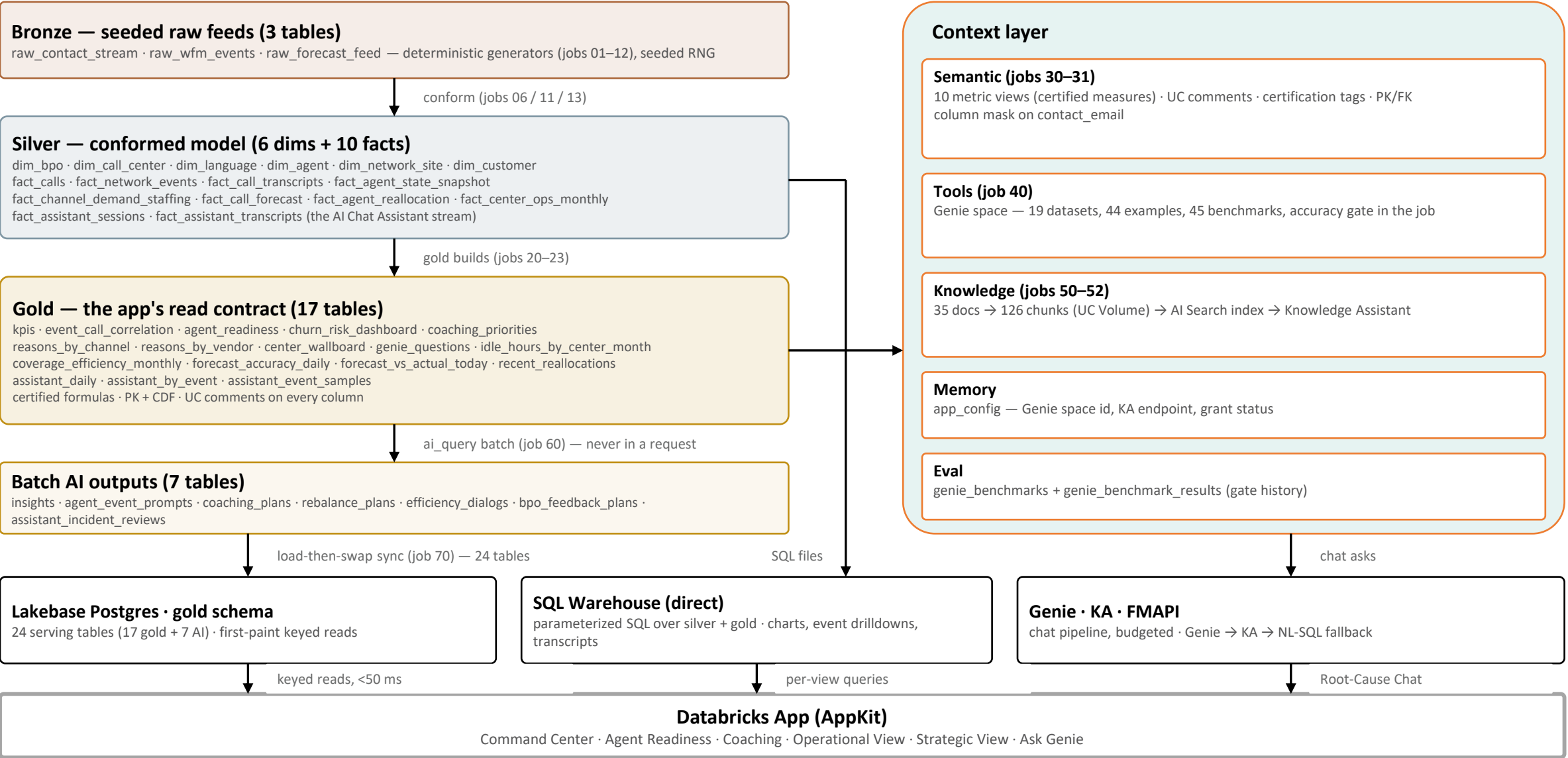


Architecture — as deployed

one install job (31 tasks) · validator green, 0 warnings
Genie gate 88.9% (40/45) · synthetic data · 3 Sep 2026



Bronze and silver — dimensions

Bronze — seeded raw feeds. Deterministic generators (seeded RNG, no wall-clock); same seed, same world.

Table	Rows	Supports
raw_contact_stream	700,220	Every voice + chat contact over 30 days, with event surges injected and attributed
raw_wfm_events	746	Simulated voice↔chat agent moves (surge, backlog, scheduled)
raw_forecast_feed	37,440	The blind baseline forecast per 30-min interval — deliberately event-unaware

Silver — dimensions. Typed, FK-verified star schema: what the warehouse queries and every gold build read.

Dimension	Rows	Supports
dim_bpo	6	Vendors (Contoso ... Northwind)
dim_call_center	8	Centres, languages, ramp dates
dim_agent	212	Roster, skills, swing eligibility
dim_language	4	EN / ES / PT / HI queues
dim_network_site	200	Network sites in 10 US regions
dim_customer	200,000	Subscriber base; masked PII column

Silver — facts

Fact	Rows	Supports
fact_calls	700,220	All contacts; voice carries event links
fact_network_events	142	Incidents; 13 active parents
fact_call_transcripts	8,000	Turn-level transcripts (1,000 calls)
fact_assistant_sessions	122,818	AI Chat Assistant sessions - its own digital stream, never in fact_calls, never a BPO
fact_assistant_transcripts	946	Sampled assistant sessions as turns (≥ 6 per active incident, worst first)
fact_channel_demand_staffing	37,440	Demand vs staffing per interval
fact_call_forecast	37,440	Forecast at the same grain
fact_agent_reallocation	746	Conformed WFM moves
fact_agent_state_snapshot	212	Live floor state (wallboard source)
fact_center_ops_monthly	89	12-month idle / coverage history

The transactional side of the star schema — thirty days of contacts, incidents, staffing and forecasts at matching grains.



Gold — the app's read contract

Table	Rows	Supports
gold_command_center_kpis	1	The seven headline tiles in one keyed read - incl. AI chat sessions today and the agent-calls + AI-chat sentiment blend
gold_event_call_correlation	101	Realized calls / sentiment / agents per incident (event drilldowns)
gold_churn_risk_dashboard	57,204	(event × customer) churn exposure — Agent Readiness at-risk lists
gold_agent_readiness	212	Per-agent readiness metrics
gold_coaching_priorities	212	Coaching queue inputs (tiers computed in-app)
gold_reasons_by_channel	23	Top contact reasons, voice vs chat
gold_center_wallboard	8	Live floor counters + languages — the vendor boxes
gold_coverage_efficiency_monthly	178	Certified coverage efficiency by centre / month / channel
gold_idle_hours_by_center_month	89	Idle-hours / occupancy story (Northwind spike)
gold_forecast_accuracy_daily	780	100-WAPE and bias by language — the degrading-forecast chart
gold_forecast_vs_actual_today	26	Today's misses + swing coverage (the 644 → 1,386 hero cell)
gold_recent_reallocations	50	Reallocation feed with event labels
gold_genie_questions	14	Starter questions; first five are the RCA chips
gold_reasons_by_vendor	138	Vendor-grain reasons - the vendor filter's re-aggregation source
gold_assistant_daily	30	AI Chat Assistant per day - sessions, containment / escalation / accuracy (from counts)
gold_assistant_by_event	13	Assistant per active incident - rates, top intent, top escalation reason
gold_assistant_event_samples	78	<= 6 sampled transcripts per incident, worst first

Pre-aggregated, certified-formula tables (PK + change feed, commented columns). All 17 sync to Lakebase; Genie answers over them too.



Batch AI outputs and the context layer

Table	Rows	Supports
insights	4	AI Executive Briefing — one narrative per topic tab
agent_event_prompts	13	Per-event agent readiness briefings (one per active incident)
coaching_plans	60	Plans for the top-60 priority queue (hash-skip on rerun)
rebalance_plans	1	The global capacity rebalance plan (Operational View)
efficiency_dialogs	1	Strategic View commentary over exactly its two charts
bpo_feedback_plans	6	Reader-to-vendor communication plan per BPO (vendor-scoped slices)
assistant_incident_reviews	13	Assistant review per incident - performance, what it got wrong, corrections, how to manage

The context layer — everything that lets AI answer correctly over gold

mv_contact_volume · mv_event_impact · mv_network_events · mv_agent_performance · mv_workforce · mv_channel_coverage · mv_reallocation · mv_forecast_accuracy · mv_assistant · mv_churn_risk
10 metric views — certified measures (Coverage Efficiency, WAPE/accuracy, bias, occupancy, exposure \$) defined once, re-derived from summed components; single-hop joins via three helper views
Genie space — 19 datasets over gold + metric views, 44 curated examples (25 pinned), 45-question benchmark suite with an accuracy gate inside the install job (currently 88.9%)
Knowledge — 35 governed docs (playbooks, SOPs, policies, the assistant playbook) chunked for AI Search, indexed by AI Search, served through the Knowledge Assistant for policy questions in chat
Memory — app_config pins the Genie space id and KA endpoint, so the app discovers its tools from data, not env drift
Governance — UC comments on every gold column, certification tags, PK/FK constraints, and a column mask on the one synthetic PII field

All AI generation is batch (ai_query on FMAPI) — the app does keyed reads with explicit not-generated states; never a model call in a request.



How it reaches the app

App view	Lakebase (first paint)	Warehouse (charts / drilldowns)
Command Center	kpis (7 tiles) · insights · assistant samples + reviews	volume, blended sentiment × severity, severity split, churn summary, drilldown (9 voice + 6 AI-chat transcripts)
Agent Readiness	churn_risk_dashboard · agent_event_prompts · assistant (by_event, incident_reviews) · At-Risk list	event selector (active incidents)
Coaching	coaching_priorities · coaching_plans	severity context for crisis adjustments
Operational View	center_wallboard · recent_reallocations · forecast_vs_actual_today · rebalance_plans · gold_assistant_daily	—
Strategic View	coverage_efficiency · forecast_accuracy · idle_hours · reasons · efficiency_dialogs · bpo_feedback_plans · gold_assistant_daily	—
Ask Genie / Root-Cause Chat	genie_questions (chips)	Genie SQL over gold + metric views; KA for knowledge; NL-SQL fallback on FMAPI

Three read paths, one rule

Lakebase sync (job 70) streams all 24 serving tables over COPY with a hard row-parity guard — the app's sub-50 ms keyed reads never touch the warehouse
Grants (job 80) give the app's service principal exactly SELECT on serving plus the UC reads the charts need; Genie and UC access are granted by the job, not by resource bindings
The chat pipeline budgets one request deadline across Genie → Knowledge Assistant → NL-SQL fallback, under the platform's ~120 s gateway ceiling

UX and theming (1-2 Sep): ? help tips on every KPI tile, predecessor icons in CNX colors, auto_awesome on every batch-AI control; chart colors from theme tokens, re-read on toggle - navy never hardcoded

The app renders precomputed answers; the only request-path model calls live inside the budgeted chat pipeline.



Archive — where it lives in Databricks

```
/Workspace/Users/herman.muir@concentrix.com/.bundle/cnx-command-center/dev/
├ files/
│   ├── databricks.yml
│   ├── src/
│   │   ├── jobs/
│   │   └── lib/
│   ├── app/
│   │   ├── client/
│   │   ├── server/
│   │   ├── config/queries/
│   │   ├── build/
│   │   └── app.yaml
│   ├── genie/serialized_space.json
│   ├── knowledge/corpus/
│   ├── metrics/
│   └── docs/
├ state/
└ artifacts/
```

← the deployed bundle source (mirror of the repo)

← the bundle: jobs, app + bindings, schemas, permissions

← the install tasks (00 schemas ... 90 validate, 95 teardown)

← contract.py story constants, generators, materialise

← React views (Command Center ... Ask Genie)

← routes + services (chat pipeline, Lakebase reads)

← the warehouse SQL files behind charts and drill-downs

← pre-built server bundle (platform builds client only)

← resource bindings: warehouse, FMAPI, job, postgres, genie

← Genie curation + benchmarks as code

← the 35 governed docs (playbooks, SOPs, policies...)

← the 10 metric-view YAML definitions

← DATA_CONTRACT.md – the integration authority

← bundle deployment state

← bundle build artifacts

Data side

catalog cnx_command_center
schemas bronze · silver · gold · context
corpus volume:
/Volumes/.../context/knowledge/corpus

Serving side

Lakebase cnx-command-center-lakebase
branch production
database databricks_postgres
schema gold — 24 synced tables

Source of truth is the git repo — the workspace copy is replaced on every bundle deploy; nothing is edited in the workspace directly.

Performance — old vs new, measured

Measure	Old	New
Questions answered (of 12)	11 — 1 refusal, 1 FABRICATED answer (reproduced 1 and 2 Sep)	12 of 12
Exact vs own ground truth (2 Sep re-run)	9 — affected-customers wrong again (a different answer every run)	11 of 12 exact + 1 governed variance (the last-week window)
Queries via certified metric views	0 of 12 (its 2 metric views unused)	11 of 12 (MEASURE() over mv_*; 8 to 9 to 11)
Measured accuracy gate	none exists	45 questions, 88.9%; install fails under 85%
Accuracy before curation (same suite)	—	52–69% (±17 pp) → curation lift +20 to +37 pp vs the current 88.9% gate
Genie latency (median)	16.7 s	16.6 s — parity (3 s polling)
Column comments on gold tables	0% (0 of 80)	97.5% (193 of 198, incl. bpo_feedback_plans)
Data freshness	window ends 18 Aug, ages daily	ends on rebuild day; refresh = one run

What changed

The failure mode, not the speed: latency is a tie; the old space fabricated an out-of-domain answer (occupancy sold as coverage efficiency, no caveat) — the new space answered everything from certified definitions

The ontology does the answering: 11 of 12 new-space queries compiled to MEASURE() over certified metric views; the old space never used its two

Question coverage roughly doubled: the WFM / forecast / coverage / language domain does not exist in the old data — and this exercise found and fixed one measure-vs-KPI convention gap (Affected Customers, mv v1.2)

Re-run 2 Sep after the fix round: new 11/12 exact - Q12 now equals the app (one coverage source); old fabricated a third time

Same 12 questions, both Genie spaces live, 31 Aug 2026, re-run 1 Sep and 2 Sep 2026; each scored against its own data. Synthetic data; latency quantized ~3–5 s by polling on both sides. Full method: old-vs-new-scorecard-2026-08-31.md.

